

Problem Set 3

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Problem 1

Consider the household savings problem:

$$\begin{aligned} \max_{\{C_t, B_{t+1}\}} \quad & \sum_{t=0}^{\infty} \beta^t u(C_t) \\ \text{s.t.} \quad & B_{t+1} = RB_t + Y_t - C_t, \end{aligned}$$

where $u(C) = \frac{C^{1-\sigma}}{1-\sigma}$, $Y_t = 0$, and B_0 is given.

1. Write down the Bellman equation.
2. Show that $V(B) = \alpha + \gamma \frac{B^{1-\sigma}}{1-\sigma}$ solves the Bellman equation and find α and γ .
3. Write down the HJB equation for the continuous-time version of the problem.
4. Solve the HJB equation using the guess-and-verify approach.

Problem 2

The profits of a firm depend on how much its (log) price p_t deviates from the optimal level normalized to zero. Each period t , the firm sets price p_t paying adjustment costs $\gamma(p_t - p_{t-1})^2$. Thus, the objective of the firm is to maximize

$$\sum_{t=1}^T \beta^t \left\{ -p_t^2 - \gamma(p_t - p_{t-1})^2 \right\},$$

taking as given p_0 .

1. What are the state variables in this problem?
2. Denote value function with V_t and solve the firm's problem in period T .
3. Solve the firm's problem in period $T - 1$.
4. Guess that the general form of the value function is $V_t = -a_t p_{t-1}^2$ and derive a recursive equation that pins down a_t .
5. Take the limit $T \rightarrow \infty$ and check whether the resulting value function solves the Bellman equation.

Problem 3

Consider the investment problem from Problem Set 1: given initial value of capital K_0 , a firm chooses investment path $\{I_t\}_{t=0}^{\infty}$ to maximize profits

$$\sum_{t=0}^{\infty} \beta^t \left[AK_t^\alpha - I_t - \frac{\phi}{2} \frac{I_t^2}{K_t} \right]$$

subject to the capital law of motion

$$K_{t+1} = (1 - \delta)K_t + I_t.$$

1. Write down the firm's problem in a recursive form.
2. Use the first-order and envelope conditions to derive the optimality condition for K_t .
3. Rewrite the Bellman equation for an arbitrary period length of Δ .
4. Derive the HJB equation taking the limit $\Delta \rightarrow 0$.